

Executive Incident Summary

On January 20, 2026, decentralized finance protocol **Makina Finance** suffered a critical exploit on its Dialectic USD (DUSD) stablecoin liquidity pool. An attacker orchestrated a complex flash-loan attack that manipulated Makina's on-chain price oracle mechanism, resulting in approximately **1,299 ETH** (roughly **\$4.1–\$5 million**) being illicitly extracted from the DUSD/USDC Curve pool in a single transaction ¹ ². The exploit was executed by **borrowing 280 million USDC** via flash loans and leveraging those funds to **distort the DUSD pool's pricing logic** ³ ². By exploiting a flaw in Makina's internal **MachineShareOracle** (the component that anchors DUSD's price), the attacker was able to withdraw far more value from the pool than should have been possible under normal conditions ⁴ ⁵. This **mechanism-level failure** – a breach of the protocol's pricing and accounting invariants – is what allowed the pool to be almost entirely drained ⁶. The incident highlights a **systemic DeFi vulnerability**: even “stable” yield vaults and liquidity pools, ostensibly designed with institutional-grade risk controls, remain susceptible to **flash loan–powered oracle manipulations**. Given Makina Finance's positioning as an advanced on-chain execution platform (with multiple audits completed in 2025), this exploit raises broader concerns about **persistent weaknesses in DeFi protocols' oracle and liquidity design**. For regulators and institutional risk managers, the Makina Finance breach underscores how a single structural flaw can **cascade into multi-million dollar losses**, threatening user funds and eroding confidence in decentralized financial infrastructure ⁷ ⁸.

Scope & Methodology

Scope: This report analyzes the technical and governance failures underlying the Makina Finance exploit, focusing on **on-chain evidence and publicly available data** from the incident timeframe. We examine the sequence of blockchain transactions involved in the attack, the smart contract functions and price oracles that failed, and the protocol's response measures. The analysis is grounded in **Ethereum mainnet data** (block and transaction records) as reported by reputable blockchain security firms and explorers, since the exploit occurred on Ethereum and centered on a Curve-based stablecoin pool ⁹ ¹⁰. We also consider official communications from Makina Finance's team and community alerts from security analysts to corroborate the on-chain findings ³ ¹¹. No private or non-public data has been used; our sources include PeckShield, CertiK, and other security researchers' real-time analyses, as well as Makina's own public statements.

Methodology: We reconstructed the incident timeline by aggregating **on-chain transaction data** and **security alerts**. Key transaction details (e.g. the flash-loan borrow, the oracle manipulation call, and the final swaps) were cross-verified with multiple independent sources: for example, PeckShield's alert on X (Twitter) provided the initial confirmation of the exploit and the attacker's fund movements ⁹ ¹², while a detailed breakdown by CertiK and others revealed the specific vulnerable function (`updateTotalAum()`) used in the attack ¹³. We traced the attacker's wallet interactions (addresses used and funds flow) via Ethereum block explorers, identifying the **block number and timestamp** of the malicious transaction (Ethereum block 24,273,362, January 20, 2026 ~03:40:35 UTC) ¹⁰. Where possible, we incorporated data from on-chain monitoring tools (e.g. TenArmor, Cyvers) that detected the event in real time ¹⁴.

Confidence Grading: Claims in this report are labeled with High, Medium, or Low confidence based on the evidence available:

- **High Confidence:** Factual details directly observable on-chain or confirmed by multiple independent security analysts. Examples include the **amount and tokens stolen (1,299 ETH)** ¹, the identification of the **vulnerable oracle function** (`updateTotalAum()`) ¹⁵, and the **addresses holding stolen funds** ⁹. These are supported by clear blockchain data and are unlikely to change.
- **Medium Confidence:** Interpretations of attacker strategy or systemic impact that are well-supported by evidence but not yet officially confirmed by the Makina team. For instance, our understanding of *how the flash loan was routed through Curve and Uniswap* to manipulate prices ¹⁶ ¹⁷, and assessments of which user groups bore the loss, rely on logical inference from on-chain behavior and analogous incidents.
- **Low Confidence:** Any speculative points (which we have minimized in this report per neutrality discipline) such as the attacker's identity or motives are **excluded** entirely. We refrain from conjecture on events not evidenced on-chain.

Exclusions: This analysis **does not assess** off-chain factors like internal team processes or any potential negligence, except insofar as governance relates to the exploit. We also do not cover subsequent market price effects on Makina's token or any trading advice – financial market commentary is outside scope by design. Importantly, we **do not speculate on the attacker's identity or any law enforcement efforts** underway, as such speculation would be neither verifiable nor relevant to the technical root cause. Our focus remains tightly on the protocol's failure mechanisms and immediate response. Finally, while we note comparisons to similar exploits, we stop short of providing prescriptive fixes or future predictions beyond the counterfactual examination explicitly requested.

On-Chain Timeline Reconstruction

• **Block 24,273,362 (2026-01-20 ~03:40:35 UTC): Initial Exploit Transaction – Flash Loan & Oracle Manipulation.** In a single atomic transaction at Ethereum block height **24,273,362**, the attacker drew a massive flash loan of **~280,000,000 USDC** from decentralized lenders (notably Aave and the Morpho protocol) ¹⁸ ¹⁹. With this instant liquidity, the attacker proceeded to interact with Makina's smart contracts. First, the attacker directed approximately **170 million USDC** into a call on Makina's internal oracle/accounting contract – identified by investigators as the `MachineShareOracle` or a related function – to **manipulate the reported price of DUSD** ⁴ ². According to CertiK's analysis, the attacker invoked a **permissionless function** `updateTotalAum()` during this transaction, which **updated Makina's total assets under management (AUM) and DUSD share price based on the now-skewed pool balances** ¹³. Crucially, this function had no time delay or governance control, so it immediately “baked in” the flash-loan-induced imbalance as the new price anchor for DUSD in that same transaction block ²⁰ (High confidence).

• **Pool Drain via DUSD/USDC Swap:** With the oracle and internal price metrics skewed in their favor, the attacker next deployed the remaining **~110 million USDC** of the flash loan to **trade against the DUSD/USDC Curve pool** ⁶. Essentially, the attacker used the manipulated price differential to execute swaps that **withdrew almost all valuable assets from the pool**. On-chain records show that roughly **\$5.1 million in USDC** flowed out of the Curve pool as a result of these swaps ¹⁷. In effect, the attacker extracted far more USDC from the pool than the DUSD they provided was actually worth under normal pricing – a direct consequence of the oracle reporting an artificially inflated DUSD value (High confidence). By the end of this sequence, the DUSD/USDC liquidity pool was **virtually emptied of stablecoin liquidity**, leaving it largely with only DUSD tokens of questionable redemption value. The

attacker's flash loan was then **fully repaid within the same transaction**, fulfilling the atomic loan condition ²¹. What remained was a net profit of assets siphoned from Makina's ecosystem.

- **Conversion to ETH and Transaction Settlement:** After draining the pool's USDC, the attacker swiftly **converted the loot into ETH** to mitigate the risk of USDC freeze or traceability. Specifically, **approximately 4.24 million USDC was swapped for 1,299.18 ETH** via a Uniswap v3 swap embedded in the same transaction ²². This yielded the attacker roughly **1,299 ETH** (worth ~\$4.13M at the time) as the final spoils of the attack. The swap and the entire flash-loan sequence executed **successfully in one block**, incurring only minimal transaction fees (indicative of the attacker leveraging an efficient bundling or block-building service) ¹⁰. The transaction completed with no errors, meaning all intermediate steps (flash borrow, oracle update, pool swap, flash repay, and ETH conversion) were executed atomically (High confidence based on blockchain transaction data).

- **MEV Bot Frontrunning and Fund Distribution:** Notably, this exploit transaction did not occur in isolation – it attracted the attention of the **Ethereum MEV (Maximal Extractable Value) infrastructure**. Blockchain observers and PeckShield noted that an **MEV builder at address 0xa6c2...** managed to **insert its own operation into the same block as the exploit** ²³ ²⁴. This MEV bot effectively *frontran* a portion of the attacker's transaction flow. However, the MEV intervention only captured a **negligible slice (~0.13 ETH)** of the profit for itself ²⁵ ²⁴, while the attacker retained the vast majority of the 1,299 ETH haul. The presence of the MEV bot underscores the competitiveness of the Ethereum mempool during such lucrative exploits – even malicious transactions face opportunistic frontrunners (High confidence, confirmed by multiple security firms).

- **Post-attack Fund Flows (Attacker Wallets):** Immediately after the exploit, the **stolen ETH** was observed consolidating into **two Ethereum addresses** under the attacker's control. According to PeckShield's on-chain alert, one address (0xbcd2...dE25) received approximately **\$3.3 million (roughly 1,040 ETH)** and a second (0x573d...910e) held about **\$880,000 (around 280 ETH)** ⁹ ¹². In total these two wallets account for essentially all of the 1,299 ETH stolen (minus the trivial MEV skim), matching the exploit's proceeds (High confidence). **No further movements** of significance have been noted from these wallets in the hours and days following the attack: as of this report, the funds have **not been sent to mixers, bridges, or exchanges** ²⁶ ²⁷. The attacker has effectively parked the stolen ETH, potentially waiting for surveillance to ease. This static behavior provides a clear trail for investigators for now, although it also raises concerns that the attacker may be hoping to negotiate or find a creative exit in the future (Medium confidence; it's an observed fact that funds are sitting, but the attacker's intent remains speculative).

- **Ancillary Observations:** On-chain security monitors (such as TenArmor and others) detected the exploit almost immediately, flagging about **\$4.2M in losses** and the nature of the oracle manipulation within minutes of execution ¹⁴. There were also tangential alerts on the Base network around the same time (e.g., concerning **Synap Logic (SYP) tokens**) which some initially suspected to be related, given the attacker's possible use of cross-chain bridges before the attack ²⁸. For instance, reports indicated the attacker's funding source included Tornado Cash on Ethereum and bridging via GasZip to Base ²⁹. However, those Base-related anomalies were later confirmed by SynapLogic as **unrelated or resolved issues** ³⁰. We mention them here only to distinguish them from the main incident: the Makina exploit itself occurred on Ethereum and was confined to the DUSD/USDC pool's mechanics, with **no direct impact on Base or other chains** (High confidence for the scope of exploit).

Exploit Vector Classification

The Makina Finance incident falls under a **flash-loan powered oracle manipulation attack**, combined with a **stable liquidity pool design flaw** in the protocol's accounting logic:

- **Oracle Manipulation:** This was the primary exploit vector. The attacker **exploited Makina's on-chain price oracle (MachineShareOracle)** by injecting a huge temporary imbalance and forcing an immediate re-calculation of DUSD's value in that distorted state ⁵ ²⁰. Because the protocol **lacked safeguards** such as time-weighted pricing or update delays, the oracle blindly accepted the attacker's manipulated input as the "true" price. This allowed the attacker to trade assets at highly favorable (false) exchange rates. In other words, the pricing **invariant that DUSD $\approx$ \$1 (backed by assets)** was broken via an oracle trick, not via a coding bug in math. We confidently classify this as an **oracle manipulation exploit** – the protocol's dependency on a **spot price feed (from its own pool balances) with no control** was the avenue for attack ²⁰ ⁷.
- **Liquidity Pool Design Flaw:** The exploit also highlights a flaw in the design of the **DUSD/USDC Curve pool integration**. Normally, Curve pools rely on arbitrage and liquidity to maintain price parity. However, Makina's DUSD pool was augmented by a custom pricing mechanism (the MachineShareOracle updating DUSD's anchor value) that could be externally triggered. The pool effectively became **vulnerable because its pricing hinged on an unprotected function**. This is a design-level issue: the pool and vault architecture did not anticipate or protect against a flash-loan induced swing. We classify this aspect as a **liquidity pool design flaw**, specifically an **invariant/accounting failure** in how the pool's balances were accounted in the vault's AUM ¹⁵ ⁷. In a robust design, no single transaction should cause a ~\$5M stablecoin pool to disgorge all liquidity; here, the invariant of pool solvency was broken by design weakness rather than a direct coding bug.
- **Vectors Ruled Out:** There is **no evidence of upgradeability or admin-key abuse** – the contracts functioned as deployed, and the attacker did not need any special permission or governance control to execute the attack. All indications are that the vulnerability was open to any user calling the function with sufficient capital, hence it was a **permissionless logic flaw** (ruled out: upgrade/permission exploit). Likewise, **governance mechanisms were not directly involved or compromised**; the attacker did not perform a malicious governance action or exploit voting power. Emergency governance features (if any) were only invoked *after* the fact by the team to pause the system, not by the attacker (thus **governance failure** is considered only in the broader sense of oversight, not as an active attack vector). Additionally, this was **not a typical smart contract bug or coding error** like a reentrancy or arithmetic overflow – it was an intended function (oracle update) used in an unintended way.
- **Uncertain/Ancillary Factors:** One subtle point is whether the exploit could be partially classified as an **economic design flaw versus pure oracle issue**. The line can blur because the oracle was part of the economic design. For completeness, we note that some analysts described it as an **"accounting/invariant failure"** since the protocol's assumption about consistent AUM was violated ⁷. However, this failure was facilitated by oracle manipulation, so we consider that the core vector. No external oracles (like Chainlink) were implicated; this was an **internal oracle logic issue**. We also rule out any **cryptographic failures** – there's no sign of private key compromise or signature forgery. All manipulations were done through legal calls to contract functions (albeit in a malicious sequence).

In summary, the exploit is best categorized as a **flash-loan oracle manipulation attack** that capitalized on a **design oversight in the DUSD stablecoin pool's pricing mechanism**. This classification is strongly supported by the facts that a rapid price swing was induced and exploited within one transaction ² ²⁰, and that a function meant to update the vault's accounting was the focal point of failure.

Root Cause Analysis

At its core, the root cause of the Makina exploit was a **failed assumption in the protocol's design**: the system assumed that updates to the vault's asset valuation (AUM) and the DUSD token's price would reflect genuine market conditions, and thus it allowed those updates to occur **immediately and without restriction**. This assumption proved false when confronted with a flash-loan attack. The specific weak point was the `updateTotalAum()` function in Makina's smart contracts, which was callable by anyone and instantly recalculated the total assets (and effectively the price of DUSD) based on current pool balances ¹³. The protocol *assumed* that no one could significantly distort those balances between blocks, or that even if they did, it would only be momentary and not coincide exactly with an AUM update. The attacker shattered this assumption by using a colossal flash loan *within one transaction* to both distort balances and invoke the update logic, thereby **tricking the system into "believing" the DUSD pool held vastly more USDC value than it really did** (High confidence in cause)

²⁰ ⁷.

Why might this assumption have seemed reasonable? Makina Finance's architecture, aimed at **institutional yield strategies**, likely prioritized up-to-the-moment accuracy of vault valuations. The ability to update AUM in real-time can be useful for reflecting yields, enabling timely deposits/withdrawals, and arbitraging across pools. Designers may have thought that **allowing permissionless AUM updates** would make the system more open or efficient (e.g., any user action could refresh the accounting, keeping DUSD's value aligned with its backing). They perhaps assumed that any price manipulation large enough to matter would be **checked by market arbitrage or by the pool's depth**. In normal conditions with moderate trades, this assumption holds: it's hard to move a well-balanced stablecoin pool by a large factor without enormous capital. However, with the advent of flash loans, **attackers do have enormous capital for one block**, and thus the assumption was **unreasonable by 2025 standards**. In hindsight, the protocol failed to account for known attack patterns where an instantaneous loan can simulate "liquidity" that vanishes the next second.

Design-level vs Implementation-level failure: This incident is best characterized as a **design-level flaw**. The smart contracts did what they were coded to do – there was no accidental bug in the code; rather, the *intended* behavior (instant oracle update from pool state) was insecure. The **absence of safeguards** (like time delay, rate limit, or multi-source price checks) in the oracle update mechanism is a design oversight ²⁰. That said, one could also view the unprotected `updateTotalAum()` as an **implementation mistake** in that it lacked an access control modifier or sanity checks. In other words, a simple implementation fix – e.g., only allow this function to be called periodically or by certain trusted keepers – might have prevented the exploit. But fundamentally, the protocol's *architecture* allowed a pool's state to directly dictate the vault's pricing in one transaction, which is an architectural vulnerability. We therefore attribute the root cause to **flawed protocol logic (architecture)** rather than a one-off coding bug.

Bug vs. Architecture Weakness: To draw that distinction clearly: a **"bug"** typically implies an error contrary to the developers' intentions (like an overflow, reentrancy that they didn't foresee, etc.). Here, the system operated as intended, but the *intention itself was insecure*. It's an **architecture weakness** – the protocol's economic design did not hold up against adversarial conditions. There was an implicit

assumption: *“The DUSD/USDC pool’s balances at any given moment can be treated as reliable inputs for price/AUM calculations.”* This assumption failed catastrophically when an attacker created an *artificial moment* that the system treated as real. The **root cause** can thus be summarized as: *Makina’s mechanism for valuing its stablecoin vault was not resilient to flash loan manipulation, due to a lack of oracle safeguards and overly permissive update functionality* ²⁰.

In simpler terms, **the protocol’s risk model did not anticipate that someone could temporarily conjure \$280M to distort prices**. This was a reasonable assumption back when flash loans didn’t exist or weren’t common, but by 2026 it was a known attack vector. The failure to integrate that knowledge into Makina’s design – despite multiple audits – points to a lapse in both design and the governance/risk oversight (see Section 7).

Response & Containment Assessment

Immediate Response and Speed: Makina Finance’s team responded to the incident on the same day, although the exploit itself was completed in seconds. In the **early hours of Jan 20, 2026**, shortly after on-chain security alerts went out, Makina acknowledged the situation via its official social media channels. The team’s initial public statement (on X/Twitter) noted: *“Early this morning we received reports regarding an incident with the \$DUSD Curve pool.”* ³¹. This acknowledgment came within hours of the attack (Medium-high confidence timing), indicating a reasonably prompt awareness. The team clarified that they were investigating and that **the issue was isolated to the DUSD liquidity pool** ³². They emphasized that **“all funds in the Machines are safe”**, meaning the underlying assets in their vaults/strategies hadn’t been compromised, and that no other strategy or asset was affected at that moment ³³. In terms of speed, this was a **timely initial response** – regulators will note that the team did not go silent or attempt to obscure the event; they confirmed it the same day. However, it’s worth noting that **no on-chain intervention** (such as a contract pause) could have mitigated the attack once it began, as it executed fully in one block. The exploit was over before any human or automated response could feasibly trigger.

Containment Measures: Upon confirming the exploit, Makina Finance **activated “security mode” (recovery mode) across all its Machines (vaults)** as a precaution ³⁴. This action likely translated to **pausing new deposits, withdrawals, or strategy operations** in those vaults while a full assessment was underway. By doing so, Makina aimed to **contain potential knock-on effects** – for example, if the attacker had found similar vulnerabilities in other vaults or if panic withdrawals might cause further issues, the system could be held in stasis. They specifically advised **liquidity providers in the DUSD Curve pool to withdraw their remaining funds** (if any) as a safety measure ³⁵. This advice indicates that **no further value could be stolen from that pool** (since it was already drained), and it was intended to prevent users from keeping capital in a broken pool that might have pricing issues until fixed. It’s a somewhat moot point since the pool was near-empty, but it shows the team trying to protect users from any additional harm (like misguided trades on the pool).

Communication Quality: Makina’s communication was factual but minimal. They **disclosed the incident and scope (DUSD pool only)** quickly ³², which is positive. They reassured stakeholders that core assets were safe and that an investigation was ongoing. However, as of the time of this report, Makina had **not yet published a detailed post-mortem or a plan for remediation/compensation** ³⁶ ³⁷. The initial statement did not describe the cause of the exploit or the extent of losses, presumably because those details were still being confirmed internally. From a regulatory perspective, the communication was **transparent in admitting a failure**, but lacked detail on *remedial steps* or accountability, which one would expect in subsequent updates. The team’s “silence” beyond the first acknowledgment was noted in some media coverage ³⁸. Given the size of the loss, stakeholders might

have expected more frequent updates. In fairness, the team likely needed time to work with auditors on root cause analysis before making further statements (a careful approach, but it leaves an information vacuum in the immediate aftermath).

Effectiveness of Technical Containment: Makina's "security mode" (pausing) appears to have been effective in preventing any further exploits during the investigation period. We did not observe any additional suspicious transactions from Makina's contracts after the mode was activated (High confidence, based on lack of further alerts). The compromised pool (being a Curve pool) could not be "paused" in the traditional sense by Makina alone, since Curve pools don't have a simple pause switch and are not solely under Makina's control. However, given the pool was essentially emptied, the threat of further draining was low. The broader platform lockdown ensured that if the attacker (or copycats) had discovered similar oracle weaknesses in other strategy pools, they could not easily exploit them immediately. It also would stop any routine operations that might inadvertently rely on now-faulty price data.

User Protection Measures: Aside from advising LPs to withdraw and urging caution, **security firms (PeckShield, ExVul, TenArmor)** externally advised users to **revoke token approvals** for Makina contracts as a safety measure ¹¹ ³⁹. This is standard practice to protect users from any potential follow-up malicious contract calls. Makina's team amplified some of these warnings (for example, by retweeting or echoing the call to revoke permissions). The team did not explicitly mention reimbursement or protection of affected users in the initial communications – no promises were made publicly about covering the losses or not, which is prudent from a legal standpoint at this early stage. However, **Makina did emphasize that underlying assets in vaults were safe**, which is effectively telling users that their deposits (outside of the pool LP positions) were intact ³³. This distinction implies **the losses fell on the pool, not the vault** – a nuance likely aimed at reassuring the majority of users that their funds were not gone.

Operational Assessment (No Moral Judgment): From an operational view, Makina Finance's response was **swift in acknowledgment and decisive in containment**. The critical system freeze (recovery mode) demonstrates they had an emergency protocol and were willing to use it. Communication was honest about the incident's existence, though details were sparse pending investigation. Areas that could be improved include more frequent updates as the situation evolved (even if just to say "investigation ongoing, no further breach, etc."), and clear guidance on next steps for affected users (e.g., will there be a formal claims process or are users simply bearing the loss?). As of now, users and regulators are waiting for Makina's full incident report and any recovery plan. The lack of an immediate **official** statement beyond social media might be viewed as a gap, but within the first 24–48 hours it's understandable as the team gathers facts.

In summary, the **incident response** can be graded as follows: **Detection/Awareness:** Good (immediate flags from community, team responded quickly). **Communication:** Moderate (transparent but minimal). **Containment:** Good (system paused appropriately). **User Guidance:** Adequate (basic warnings given, but awaiting more info). There is no indication of any **cover-up or misinformation** – Makina acted in a generally responsible manner post-incident, with the caveat that the true test will be how they follow through with a detailed analysis and any compensation or fixes in the coming days.

Governance & Control Surface Review

The Makina Finance exploit also prompts a review of the protocol's **governance and administrative controls**, as these can either mitigate or exacerbate such incidents:

• **Admin Keys and Privileges:** Makina's architecture includes governance and admin components – the documentation references a **Governance module managing permissions, risk parameters, and updates** ⁴⁰ ⁴¹ . This implies that privileged roles (likely a multi-signature admin wallet or a DAO) have the ability to upgrade contracts, adjust settings (like oracle parameters), or pause functionality. In practice, after the exploit, the team was able to **invoke a “recovery mode”** across vaults, which suggests they had an admin capability or emergency switch coded in. It's likely this power resides with the core team's multi-sig or the designated DAO, rather than an autonomous code trigger. Such admin keys are a double-edged sword: they allow quick intervention (which was beneficial here to halt operations), but they also represent a **centralized point of trust** and, if compromised or misused, a potential systemic risk. In Makina's case, there is no evidence of malicious admin use – **the attacker did not target admin keys** and the team's use of admin powers post-incident was for stabilization. Regulators should note, however, that **the existence of these admin controls means Makina is not fully decentralized in emergency handling**, which is common in DeFi but relevant for assessing who ultimately holds power over user funds.

• **Upgradeability and Smart Contract Governance:** Makina's contracts underwent multiple audits in 2025, and the presence of *Core* and *Periphery* modules suggests an upgradeable or modular design ⁴² . Typically, protocols have a timelock and/or governance vote required to upgrade critical contracts. We do not have explicit data on Makina's timelock or whether the governance is token-based or multisig-based. However, given the speed of response (toggling recovery mode quickly), it suggests a **multisig-controlled admin** rather than a slow token governance process. One governance question raised by this exploit is: *Why was the `updateTotalAum()` function permissionless and not restricted by governance or an owner?* That decision itself is a governance design choice (to allow open access for efficiency). In hindsight, it might have been a misjudgment. A robust governance process might have flagged the risk in that design and either gated it (only the strategy controller or an authorized keeper can call it) or required a delay. If Makina had a risk committee or security council, this function could have been identified as high-risk. It's unclear if such internal review happened or if governance forums discussed oracle risks pre-launch – likely not in depth, given it remained exploitable.

• **Emergency Powers and Their Use:** We have seen Makina's team use emergency powers to pause operations (security mode). There is no indication they have, or attempted, any ability to reverse transactions or recover funds via governance (no such thing exists on Ethereum without forking the chain, which is not feasible for an isolated DeFi hack). They did not have an automated circuit-breaker that stopped the attack mid-way – because the design didn't include one and the attack concluded in one block. In retrospect, a possible *governance-controlled circuit breaker* (for example, a parameter capping how much DUSD price can move per transaction or a pause if pool imbalance exceeds X) could have mitigated this, but implementing such requires foresight. The incident reveals that **Makina's governance had not preemptively deployed any circuit-breakers on the DUSD pool oracle** – a governance oversight. After the fact, emergency governance was used appropriately to prevent further damage. There's no sign of delay or internal bickering; the team acted decisively, indicating a **centralized emergency process** (which can be good for containment, though again it underscores centralization).

• **Transparency of Authority:** From an external perspective, **it is not fully transparent who holds Makina's admin keys or how their governance is structured**. The protocol launched Season 1 recently (assuming by context and docs) and is likely in a semi-centralized state with core contributors in control. They have a token (\$MAK), but how much governance that token has versus team multi-sigs is not immediately clear. For risk assessment, we assume the team (or the Operator, Dialectic) had significant control. There haven't been public statements detailing the governance voting on the emergency response, suggesting it was an internal decision by authorized persons, not a token-holder vote. This is typical for crisis handling in DeFi's early stages but should be noted.

• **Implications for Systemic Risk:** The exploit demonstrates a **latent risk in Makina's control surface**: the combination of a powerful admin structure and a complex strategy engine means that *if* an attacker had compromised the admin keys or governance, the damage could be even worse (they could rug entire vaults, etc.). That did *not* happen here, but it's worth asking if Makina's governance process is robust enough. Additionally, since the exploit was due to a design flaw, one must consider **governance's role in risk management** – was this a failure of internal risk governance? Did the project's control framework (which might include code audits, economic audits, etc.) miss something known in the wider community? For regulators, the takeaway is that **even well-audited, institution-focused DeFi projects can harbor hidden risks** that governance processes did not catch.

The **systemic risk** beyond Makina itself is that similar oracle designs might exist in other protocols – something for the broader governance community to investigate. Makina's incident may spur its governance to implement more safeguards (like requiring oracle updates to use time-weighted prices or multi-tx consensus). It also raises the question of **transparency**: will Makina disclose the specifics of their admin key setup (e.g., multi-sig addresses, number of signers) and any compensation plans via governance votes? Doing so would improve trust. Right now, trust in Makina's governance likely took a hit because the exploit hints at insufficient risk oversight.

In summary, **Makina's governance and control framework allowed quick emergency action but failed to prevent the exploit**. The lack of an oracle safeguard in code points to a lapse in the risk management aspect of governance. The usage of admin powers post-hack was appropriate, but it serves as a reminder that **users ultimately rely on the team's discretionary control in emergencies**, which is a concentration of authority that regulators will scrutinize. Moving forward, one would expect Makina governance to institute changes addressing the oracle design, and to increase transparency about how decisions are made to restore user confidence.

Economic Impact & Trust Damage

Direct Economic Losses: Approximately **\$4.13 – \$5 million in value was illicitly extracted** from Makina Finance's ecosystem during this exploit ¹ ². In concrete terms, the attacker withdrew about **1,299 ETH** from the protocol (after converting stolen USDC to ETH) ¹. This sum represents the **loss of liquidity from the DUSD/USDC pool** – effectively, the pool's USDC reserves were taken. It's important to clarify where this money came from: the DUSD/USDC Curve pool had roughly \$5M in total liquidity prior to the attack (split between DUSD and USDC). That pool was largely or entirely drained, meaning those funds are the ones now sitting in the attacker's wallets. The **Total Value Locked (TVL)** in Makina's DUSD strategy and related pool thus dropped by about \$5M immediately. If Makina's overall platform managed, say, tens of millions across all strategies, this is a significant chunk. (For context, DUSD's circulating supply was on the order of ~\$50M before the attack, per public data, which suggests roughly 10% of that value was in the Curve pool and got siphoned.)

Affected User Classes: The primary victims are the **liquidity providers of the DUSD/USDC pool**. These likely included some combination of **Dialectic (the strategy operator)**, possibly Makina's own treasury or seed liquidity, and other users who provided either DUSD or USDC to that Curve pool to earn fees. Those LPs now hold near-worthless pool tokens: the pool's USDC side is gone, and they're left mostly with DUSD (or an imbalanced position) that cannot be easily swapped back to real USDC at parity. If Dialectic or Makina itself was the main liquidity provider (a common scenario where protocol sponsors their pool), then the protocol's treasury or the operator fund absorbed much of this loss. If external users were LPs, those users lost funds. **DUSD holders** (i.e. users who deposited into the DUSD vault but did not provide liquidity) were *indirectly* affected in that their means of swapping out of DUSD was disrupted. However, Makina claims the underlying assets for DUSD are safe in the vaults ³³, so in

theory DUSD holders can still redeem through the vault's normal process (though perhaps with delays and now without the convenience of the Curve pool).

It remains to be seen if the vault itself is undercollateralized due to the exploit – if the attacker minted DUSD illegitimately, that could dilute other holders. But based on Makina's statement that "all funds in the Machines are safe," it suggests **the vault's assets (like deployed USDC, etc.) remain intact**, and the loss is confined to the pool LPs. In either case, **trust among those users is undermined**: LPs have tangible losses, and DUSD holders have lost a key liquidity outlet and likely confidence in the 1:1 stability of DUSD.

TVL and Ecosystem Impact: Beyond the immediate \$5M, the psychological impact on Makina's TVL could be larger. **Users may withdraw funds from other Makina strategies as a precaution**, lowering TVL further. Even though other "Machines" were not exploited, a logical response for risk-averse users is to pull capital until Makina proves the issues are fixed. This kind of incident often triggers a drop in protocol TVL across the board as trust is reevaluated. If Makina had, hypothetically, \$100M in various vaults, some of that is now at risk of exiting, which could also force those vaults to unwind positions. Such second-order effects can be significant: e.g., if DUSD vault's assets are deployed across DeFi, a sudden withdrawal wave might necessitate asset sales, potentially causing slippage or minor losses in those strategies.

Impact on Dialectic and Institutional Partners: Notably, DUSD is described as Dialectic's yield-bearing dollar product ⁴³. Dialectic is presumably an institutional asset manager linked with Makina. This exploit not only costs money but also **damages Dialectic's reputation as a secure operator**. Institutional partners or clients of Dialectic will be concerned that a strategy Dialectic runs had such a flaw. Even if those partners didn't lose funds (perhaps Dialectic might choose to make LPs whole if they were clients), the trust in Dialectic's "robust framework" ⁴⁴ is dented. For Makina, which markets itself as an "institutional-grade execution engine," this event is a blow to that branding.

User and Market Confidence: From a market perspective, any governance token Makina has (e.g., \$MAK) could see a loss of confidence. (While we won't discuss price, we note generally that hacks of this magnitude often correlate with a drop in the protocol's token value, reflecting shaken confidence). More qualitatively, **existing users may hesitate to deposit fresh funds**, and prospective users (especially institutional ones) could reconsider using Makina for yield strategies. Competing platforms might gain an edge if they can claim stronger safety.

Second-order Effects: Systemically, this exploit adds to a growing list of DeFi hacks, which in 2025 reportedly saw over \$3.4B stolen ⁴⁵. Each incident marginally erodes general trust in DeFi for outsiders. For regular DeFi participants, the Makina hack reinforces the "*no platform is safe*" sentiment – even a project with multiple audits by reputable firms (SigmaPrime, ChainSecurity, Ottersec, etc. per Makina's docs ⁴²) can suffer a critical failure. This could lead to increased demand for **insurance products** (though insurers themselves often exclude oracle-manipulation claims or have caps). If Makina had any insured funds or backstops, this will test them.

Within the Makina community, there may be contentious discussions: should the protocol **socialize the losses** (e.g., via a DAO fund if one exists) to compensate LPs? Or is it solely users' risk? Such decisions can further impact trust. If users are left with losses and no compensation, some will exit permanently. If the team or investors step in to cover losses, that could restore some confidence but at financial cost.

In summary, the economic impact of ~\$4-5M is material but not massive in absolute DeFi terms (we've seen larger hacks). However, the **qualitative trust damage is significant** given Makina's target

audience. The incident underscores that even sophisticated platforms carry smart contract risk. For regulators, it exemplifies how losses in a single pool can propagate into **broader credibility issues** – users lose faith not just in one pool but in the protocol’s safety culture. Rebuilding that trust will require transparency, fixes, and possibly user compensation, none of which have yet occurred as of this report. Until then, Makina faces a trust deficit: a regulator or institutional risk officer would likely impose higher risk premiums on Makina-related activities or reduce exposure to it in the short term.

Counterfactual Analysis

Could this exploit have been prevented? Yes – in hindsight the attack was **preventable with standard defensive design**. There are a few key counterfactual conditions that, had they been in place, would have averted or dramatically mitigated this incident:

- **Oracle Update Safeguards:** If Makina’s `MachineShareOracle` (or `updateTotalAum()` function) had been designed with **any kind of delay or averaging mechanism**, the attacker’s instantaneous price manipulation would not have registered as a valid price. For example, a **time-weighted average price (TWAP)** over even a few minutes, or a requirement that price updates be derived from multiple blocks’ data, would have meant the flash-loan spike was ignored or smoothed out ²⁰ ⁷. Similarly, a **circuit breaker** that detects a rapid change in reported AUM or pool ratios could have paused new oracle updates. These techniques are known in the industry: many protocols use them explicitly to guard against flash-loan manipulation. Makina not employing them was a choice – presumably to allow real-time pricing – but a choice that left the door open to attack. In a counterfactual scenario where Makina followed **industry-standard oracle design** (e.g., using Chainlink or a guarded launch for their in-house oracle), this specific attack would likely have failed.
- **Permissioned Oracle Updates or Access Control:** Another preventive measure could have been restricting who can call the sensitive `updateTotalAum()` function. If only a **trusted keeper or the contract itself** could trigger AUM recalculation (for instance, at set intervals or after certain actions), an attacker couldn’t have arbitrarily invoked it mid-transaction ¹⁵. The fact that *anyone* could call it was a design choice that prioritized openness over safety. Had Makina required, say, a governance vote or multi-sig sign-off to adjust the core price anchor (clearly not practical for frequent updates, but perhaps a smaller design tweak like only allowing it at the end of a block or only once per hour), the attacker’s one-shot manipulation would be much harder. In summary: **stronger access control on critical state-changing functions** is a known safeguard that was absent here.
- **Precedents and Warnings:** The type of attack used on Makina is not new; **similar flash-loan oracle exploits have been well-documented since at least 2020-2021** (e.g., Harvest Finance’s 2020 hack, and many in 2022 as discussed in Section 10). There have been numerous post-mortems in the DeFi community emphasizing time-weighted or independent oracles for pricing vault shares. Makina’s multiple audits in 2025 **should have caught this risk** – and indeed it’s possible auditors noted it but perhaps the team believed their specific implementation was safe or the pool was “too large to manipulate”. If so, that was a misjudgment. Counterfactually, if audit or internal risk assessments had given more weight to this scenario (perhaps performing a simulation of a flash-loan on the Curve pool), the vulnerability could have been identified and patched pre-launch. The tools to do so (fuzzing, invariant testing – which Makina did with Enigma’s audit ⁴² – could have been applied to price manipulation scenarios). So one might say this exploit could have been prevented by **more thorough adversarial testing and by heeding**

known precedents (Medium confidence; we infer from similar cases that the issue was knowable).

- **Industry Safeguards – Standard vs Below Standard:** Relative to industry standards in late 2025, Makina’s safeguards here were **below standard**. Many DeFi protocols learned from earlier incidents to not trust raw pool prices in a single block. For instance, stablecoin issuers or vaults often implement **sanity checks** (e.g., not updating if price moves beyond X% in one block, or requiring an off-chain sign-off for large swings). Makina’s design omitted these, placing them behind the curve. It’s worth noting some recent exploits (like the **Moonwell exploit in Nov 2025**) similarly highlighted oracle issues, showing not all protocols adopted best practices ⁷ ⁴⁶. But the trend is that *flash-loan resistance is expected*. So yes, the defense was below the evolving industry standard.

In a counterfactual world where Makina had adopted any of the above measures, the attacker’s job would have been much harder, if not impossible. For example, if a TWAP of even 10 minutes was required, the attacker would need to sustain the manipulation across multiple blocks, which is far more expensive and likely unprofitable (arbitragers could step in during that window). Or if the function was not publicly callable, the attack as executed simply couldn’t happen; the attacker would be stuck with a transient imbalance that the system doesn’t lock in, and they’d repay their flash loan with nothing gained.

Were there known precedents? Absolutely. As noted, **oracle manipulation via flash loans** is a notorious vector. The DeFi community has seen it in **Mango Markets (Oct 2022, \$117M)** ⁴⁷ ⁴⁸, **Inverse Finance (multiple times in 2022)**, **Harvest Finance (2020)**, and others. In many ways, the Makina exploit is a textbook case: borrow large flash loan, manipulate on-chain price, extract value, repay loan. The fact that it succeeded in 2026 suggests either complacency or a belief that “our case is different.” It wasn’t. So yes, precedents existed and should have informed Makina’s design. Indeed, Makina’s own auditors or advisors likely were aware of these. This raises a governance question: were these safeguards consciously decided against (for performance or complexity reasons), or simply overlooked? We cannot say, but the outcome shows a **missed opportunity to learn from prior incidents**.

In conclusion, **this exploit was preventable**. No novel zero-day bug was needed – just a combination of lots of capital and a known trick. Proper oracle design and risk controls could have thwarted it. One can argue that by 2026, *not* implementing such safeguards is a lapse in duty of care for a DeFi protocol marketing itself to institutions. While that is a harsh judgment, it aligns with the fact that countless warnings in the ecosystem highlight exactly this kind of risk ⁷. The counterfactual is clear: **with even minimal anti-flash-loan oracle protections, Makina’s DUSD pool exploit would have been either impossible or far less severe**.

Comparative Precedent Analysis

The Makina Finance exploit shares characteristics with several historical DeFi exploits, underlining recurring themes in protocol failures:

- **Mango Markets (October 2022, Solana):** Mango’s case is one of the most famous **oracle manipulation attacks**. There, the attacker (Avraham Eisenberg) **inflated the price of Mango’s low-liquidity governance token (MNGO)** by using capital across exchanges, and then used the overvalued MNGO as collateral to borrow all available assets from the platform ⁴⁷ ⁴⁸. The result was ~\$117M in various crypto drained. The similarity to Makina: both involved *deliberately*

mispricing an asset to withdraw value. In Mango, the asset was a governance token used in collateral; in Makina, it was a vault token (DUSD) used in a pool. Both attacks relied on there being **insufficient safeguards on price inputs**. Mango's attacker infamously argued his actions were just a "profitable trading strategy" exploiting the protocol as designed ⁴⁹ – essentially the same could be said here: Makina's code allowed it, so the attacker didn't need to hack the code, only game the economics. Mango's outcome was interesting: the attacker negotiated and **returned a portion of the funds** after a DAO vote, but was later arrested and charged with market manipulation ⁵⁰ ⁵¹. *Recovery outcome:* Mango users eventually recovered some funds (about \$67M of the \$117M after negotiations), but the incident led to major governance turmoil and legal consequences. For Makina, it's too early to know if a similar negotiation or partial return will occur; however, Makina's attacker being relatively silent (so far) and using Tornado suggests a less cooperative stance. The key lesson from Mango – enforce robust price oracles – echoes in Makina's case.

- **Harvest Finance (October 2020, Ethereum):** Harvest, a yield aggregator, suffered a ~\$34M exploit when an attacker used a flash loan to **manipulate the price of stablecoins in a Curve Y pool**, tricking Harvest's vault into overvaluing a deposit and allowing the thief to withdraw more stablecoins than deposited. This is an extremely close analog to Makina's situation: a Curve stablecoin pool was manipulated, and the vault didn't have a TWAP on the price. Harvest's attacker quickly laundered funds through renBTC etc. Harvest ended up partially compensating users from a treasury fund but lost significant trust. The pattern is the same – **flash loan + Curve pool imbalance + vault misvaluation**. One difference: Harvest's case spurred many projects to add time-weighted or delayed price feeds for vault strategies. Makina's design should have benefited from those industry discussions, yet it fell to essentially a variant of the Harvest trick five years later.
- **Inverse Finance (April 2022 & June 2022, Ethereum):** Inverse had two oracle-related exploits. In April, an attacker manipulated the price of a token (XIM) on a money market to borrow ~\$15M of assets. In June, a flash loan was used to inflate the price of collateral (YFI in a thin market) and drain ~\$1.2M ⁵². Inverse's incidents underscore how **reusing known oracle patterns without guardrails leads to repeat exploits**. Inverse attempted to compensate some users and tighten oracle sources after the first hack, but was still exploited again. This is cautionary for Makina: a single fix (patching one function) may not be enough if the overall oracle approach isn't reconsidered.
- **Moonwell (November 2025, Base/Optimism):** A very recent case, Moonwell, lost ~\$1M when an **off-chain oracle feed (for a staked ETH derivative) malfunctioned** and reported a wildly inflated price ⁵³ ⁵⁴. Attackers flash-loaned minimal collateral to borrow large amounts, similar in concept to Makina though via an external oracle failure. Moonwell's case underlines that whether the oracle is internal (as in Makina) or external, **the DeFi protocol must have checks for plausibility**. Moonwell had a series of exploits, showing that partial fixes without holistic review left them vulnerable repeatedly ⁵⁵ ⁵⁶. The parallel: Makina must thoroughly audit all strategies for similar issues, not just patch DUSD, else it risks a series of "whack-a-mole" exploits.
- **Poltergeist/Polter Finance (Nov 2024):** Less known but Polter Finance was hit by an ~\$8.7M price manipulation exploit on its stablecoin pools, forcing them to pause and eventually shut down that product ⁵⁷. It was another case of an **invariant assumption broken by a flash-loan driven price swing**. Polter's team paused their contracts and prevented a larger loss. Comparatively, Makina's loss was in the same ballpark (single-digit millions). Polter's recovery involved absorbing losses and compensating via new tokens – essentially a painful path that Makina might have to consider if they cannot cover losses outright.

Failure Similarities: All these incidents share a core failure: *the protocol relied on a price that was assumed to be fair or “real,” and attackers showed it could be artificially moved.* **Whether it’s a governance token, LP token, or vault share, if its valuation can be manipulated in a short window, the protocol’s accounting can be exploited. Flash loans and MEV techniques are common threads enabling these short-term manipulations. Another similarity is that most of these protocols had audits and sophisticated developers**, yet the exploits were often not low-level bugs but economic design oversights** – a reminder that audits must include economic modeling, not just code review.

Differences and Outcomes: A difference often lies in *response and outcome*. For example, Mango’s situation involved the attacker negotiating openly (an extreme rarity) and law enforcement stepping in, whereas in something like Harvest the attacker remained anonymous and no funds were recovered except what the dev team could muster. At this stage, Makina’s attacker has not surfaced to negotiate, and given the relatively smaller amount and anonymity tools used, we might expect a **non-recovery outcome** (though we do not predict, we can say historically most such attackers do not return funds unless there’s strong incentive or pressure). Some protocols managed to stay afloat after similar hacks (Harvest, Inverse, etc., though they lost market share), while others faced existential crises (if users flee en masse). Makina’s future will depend on how they address the exploit and reassure users.

From a **regulator’s view**, the recurrent pattern is clear: these failures aren’t one-off freak events but rather part of a **systemic pattern in DeFi** where innovative financial engineering sometimes repeats mistakes in new forms. The Makina exploit is another entry in that ledger, comparable to Mango Markets’ high-profile case in showing that such manipulation is considered illegal market manipulation by regulators ⁴⁹. This context might influence Makina’s decision-making – for instance, cooperating with investigations or pursuing the attacker, as Mango’s DAO did by suing their exploiter ⁵⁸.

In conclusion, **Makina’s exploit is not an isolated anomaly**; it fits a known class of DeFi attack vectors. The comparisons to Mango, Harvest, and others highlight that the DeFi industry has learned these lessons before – though sometimes unevenly applied. The outcomes historically range from partial recovery to none, with longer-term reputational damage. Makina can learn from how those projects responded: transparency, swift code fixes, and where possible compensating users or involving law enforcement have been parts of successful recoveries (or at least mitigations). But no recovery should be assumed; in many cases users bear the loss, and the protocol’s survival hinges on restoring trust and demonstrating that the root cause is decisively fixed going forward.

Open Questions & Unknowns

Despite the information gathered so far, several critical questions remain **unanswered or uncertain** at this time:

- **Loss Attribution & Compensation:** *Who* ultimately bears the ~\$4.1M loss from the DUSD/USDC pool? It’s unclear how much of the pool was funded by **user liquidity providers vs. the protocol’s own treasury or partners**. Depending on this, the impact is either dispersed among users or concentrated on Makina/Dialectic. Correspondingly, it’s unknown whether Makina **plans to compensate affected LPs** or if those losses are considered final. No official statement on reimbursement or an insurance claim (if any) has been made yet.
- **Root Cause Confirmation:** While external analysis points to `updateTotalAum()` and oracle manipulation as the culprit, we lack an **official post-mortem from Makina** confirming the precise vulnerability. There might be nuances (e.g., was there a price feed from the Curve pool directly, or was it entirely internal logic?) that only the team’s report can clarify. The *exact*

mechanism of how DUSD was mis-priced – for instance, did the attacker mint extra DUSD or simply trade on a false price – remains partly speculative without the team's detailed breakdown.

- **Existence of Broader Vulnerabilities:** Are similar flaws present in other “Machines” or strategies on Makina? The DUSD strategy was one product; Makina runs others (perhaps different vaults for other assets). It's unknown if the **team has audited all other strategies** for analogous oracle or accounting issues. Until Makina discloses the results of a comprehensive review, we cannot conclude that this exploit was isolated. There's a latent concern: could any vault that relies on on-chain prices be manipulated similarly? This open question will persist until code fixes or audit reports are released.
- **Attacker Identity and Intent:** The identity of the attacker is unknown. They used Tornado Cash and complex routing to obfuscate origin ²⁹. We do not know if this was an external opportunist or someone with insider knowledge of Makina's design (given how targeted the attack was). There's no indication the attacker has communicated (e.g., no messages or demands were embedded in transactions as sometimes happens). **Motives** beyond financial gain (e.g., ethical hacking vs malicious) are also unknown. This matters for whether negotiation is possible. At present, the attacker's profile remains a black box.
- **Law Enforcement or Regulatory Actions:** It's unclear if Makina has engaged with law enforcement or if any investigation is ongoing beyond on-chain tracing. Given the precedent with Mango (where U.S. agencies got involved), one question is whether authorities will treat this as a market manipulation case. As of now, no public information on subpoenas or investigations has surfaced. Makina's jurisdiction, the affected users' jurisdictions, and whether this falls under any specific regulatory purview (e.g., commodities manipulation if DUSD is viewed as such) are all open questions.
- **Makina's Recovery Plan:** Beyond technical fixes, what is Makina's plan to restore confidence? Will they publish a full audit of what happened and an actionable roadmap to prevent future incidents? Will they invite third-party auditors to re-review the updated code? The **timeline for resuming normal operations** is unknown – the Machines are in “recovery mode” indefinitely until the team concludes their investigation. Users and observers do not yet know how long that will take or what criteria will signal that Makina is safe to use again.
- **Treasury and Financial Health:** We do not know Makina's financial reserves or ability to absorb the loss. If Makina or Dialectic decides (or is expected) to cover the stolen \$4M, can they do so without jeopardizing their solvency? The *financial health of the project post-incident* is an open question. Any attempt to raise emergency funds or use treasury assets (if governed by token vote) has not been announced.
- **Disclosure of Governance and Control:** This incident may prompt questions about **who controls Makina** (keys, governance) – it's unknown if the team will disclose more about their admin multisig or add additional security measures (like adding independent overseers or time-locking critical functions). As of now, the governance processes remain opaque to outsiders, and it's uncertain if that will change.
- **Future Regulatory Classification:** As regulators examine this incident, an open question is whether **DUSD or similar vault tokens will face scrutiny as potentially needing regulation** (e.g., as unregistered securities or commodity pools) given the manipulation. While speculative,

it's unknown how this exploit might feed into regulatory narratives. No immediate regulatory response has been noted, but it's something to watch.

In summary, much remains unknown pending Makina Finance's forthcoming actions and disclosures.

Key data gaps include:

- The **full technical post-mortem** from Makina's team (to confirm all details of the exploit path).
- Any **compensation plan** or user remediation approach.
- Assurance that **other strategies are safe**, which only an audit or code review can provide.
- Information on whether external entities (authorities or insurers) are involved in the response.

Until these unknowns are addressed, our analysis must be caveated as based on the best available public information, which, while substantial about the exploit mechanics, does not cover everything about the aftermath.

Disclosure & Neutrality Statement

The author of this report discloses that the commissioning entity holds a **minor token exposure (less than USD \$5,000)** to Makina Finance. This fact has been explicitly noted to uphold transparency. **No compensation was received** from Makina Finance or any affiliated party for conducting this analysis, and there has been **no collaboration or prior coordination with the Makina team** in preparing these findings. The analysis has been carried out independently, with the sole aim of informing public stakeholders (regulators, risk committees, auditors, and policy evaluators) about the incident's facts and implications.

Every effort has been made to maintain strict neutrality and objectivity. The minor financial exposure mentioned is assessed as immaterial to the conclusions and has not influenced the analysis or tone of the report. In line with bias-mitigation discipline, the report avoids any advocacy on behalf of Makina or any other entity, and it refrains from speculative or promotional statements. We have **defaulted to conservative, evidence-based interpretations** throughout. The findings focus on technical and governance factors without offering investment advice or moral judgment.

This report is provided in the **public interest**, aiming to support informed decision-making and oversight. All conclusions are drawn from evidence and are presented without favor or prejudice.

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